The exam contains 12 questions. All questions ask for an evaluation of five statements with (yes/no) or (true/false) answers. A true answer adds +1 points and a false answer adds -1 points to your exam score. Hence, skipping to evaluate a statement is very likely to be more advantageous for you than making a random guess. Your final score is calculated as (Your point sum)/(12*5)*100. The exam duration is four hours.

$P(\cdot)$ denotes a probability measure. $\mathbb{E}[\cdot]$ denotes expectation. $\mathrm{Var}[\cdot]$ denotes variance.

The symbol $\top$ denotes matrix or column vector transpose.

No assumptions may be made additional to those specified in the question text.

Question 1

Consider the supervised learning problem of predicting the real-valued scalar y from a feature vector x to be solved by fitting a ridge regression predictor

$$L(w) = \frac{1}{n}\sum_{i=1}^{n}(y_i - w^\top x_i)^2 + \alpha||w||_2^2$$

to a data set $D = \{(x_i, y_i) : i = 1, \dots, n\}$ where w is a parameter vector with appropriate size and a positive constant α. Evaluate the statements below.

Remarks: Model variance is defined with respect to the experiment repetitions. It is the variance that is used in the context of the bias-variance dilemma.

Answer 1.1: Model bias increases as n increases.

Answer 1.2: Model variance increases as α increases.

<u>Answer 1.3:</u> The overfitting risk of the model reduces as α increases.

Answer 1.4: Model capacity increases as α increases.

<u>Answer 1.5:</u> The underfitting risk of the model reduces as α decreases.

Question 2

Consider the confusion matrix below obtained from a classifier evaluated on the test split of a data set comprising data points with discrete labels that can take three possible values: A, B, C. Each of these values corresponds to a class.

		Prediction A	B	C	
	A	15	5	5	25
Actual	B	10	20	5	35
	C	5	10	25	40
		30	35	35	100

Evaluate the below statements regarding this confusion matrix.

Answer 2.1: The precision for Class A is greater than the recall of Class A.

<u>Answer 2.2:</u> The precision for Class B is equal to the recall of Class B.

<u>Answer 2.3:</u> The precision for Class C is greater than the recall of Class C.

<u>Answer 2.4:</u> The recall for Class A is greater than the recall for Class B.

Answer 2.5: The accuracy of the classifier is smaller than 0.5.

Question 3

Consider a coin with a heads probability of 2/3 being tossed three times. Denote by the random variable C the number of times the coin toss results in a heads. Evaluate the following statements related to the random variable C.

Answer 3.1: $P(C = 1) > P(C = 3)$

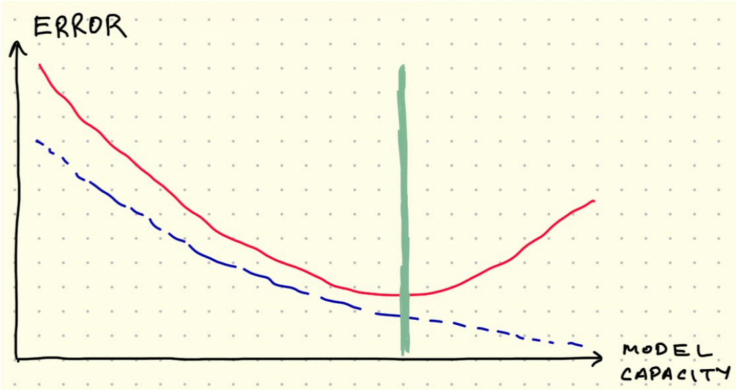
Answer 3.2: $\mathbb{E}[C] > 2$

<u>Answer 3.3:</u> $\mathbb{E}[C^2] > \mathbb{E}[C]^2$

<u>Answer 3.4:</u> $P(C = 0) < P(C = 3)$

<u>Answer 3.5:</u> $\mathbb{E}[C^2] > \mathrm{Var}[C]$

Question 4



Evaluate the following statements about the figure above.

Answer 4.1: The solid red line shows the model prediction error on the test split.

Answer 4.2: The dashed blue line shows the model prediction error on the training split.

Answer 4.3: The right-hand side of the solid vertical green line indicates the underfitting region.

Answer 4.4: The optimal model capacity is marked by the green line.

Answer 4.5: Doubling the size of the training data without changing the size of the test data would shift the vertical green line to the left.

Question 5

Suppose A and B are events with $0 < P(A) < 1$ and $0 < P(B) < 1$. Evaluate the statements regarding these events.

Answer 5.1: If A and B are disjoint, they cannot be independent.

Answer 5.2: If A and B are independent, they cannot be disjoint.

Answer 5.3: If $A \subset B$, then it is possible that A and B are independent.

Answer 5.4: If A and B are independent, then it is possible that A and $A \cup B$ are independent.

Answer 5.5: There exists A and B that satisfy $P(A \cup B) = 1$.

Question 6

Evaluate the following statements about the $k-$means clustering algorithm

Answer 6.1: The algorithm always converges to the same cluster centroids regardless of its initialization.

Answer 6.2: The algorithm can never converge to a clustering before changing the assignment of at least one data point at least once after initialization.

Answer 6.3: The computational complexity of the algorithm is not dependent on the data dimensionality.

<u>Answer 6.4:</u> The algorithm assigns each data point to a single cluster.

<u>Answer 6.5:</u> The algorithm cannot output spiral-shaped clusters when $k > 1$ and the algorithm is run until convergence starting from a random initialization.

Question 7

Consider a predictor that maps a $d-$dimensional input vector x to an output prediction with

$$f(x) = W_2^\top \max(W_1^\top x, 0)$$

for $W_1 \in \mathbb{R}^{d\times h}$ and $W_2 \in \mathbb{R}^{h\times 1}$ and the $\max(\cdot)$ operator applies element-wise to the vector on its argument.

Answer 7.1: The predictor can reach zero training error only if the input-output relationship is perfectly linear, i.e. there exists a vector $\beta \in \mathbb{R}^d$ such that $y_i = \beta^\top x_i$ for all training inputs x_i and the corresponding output labels y_i.

<u>Answer 7.2:</u> The predictor corresponds to a neural network with a single hidden layer of size h.

<u>Answer 7.3:</u> The predictor corresponds to a neural network that uses a rectified linear unit as its activation function.

Answer 7.4: The predictor cannot map an input outside the $[0, 1]$ interval no matter which values W_1 and W_2 take.

Answer 7.5: The gradient descent algorithm is guaranteed to converge to the global minimum of the loss function $(y - f(x))^2$ if the learning rate is chosen small enough.

Question 8

Evaluate the statements below regarding a data set $D = \{X_i : i = 1, \dots, n\}$ that contains n observations X_i that come independently and identically distributed from a Bernoulli distribution with parameter π.

Answer 8.1: The maximum likelihood estimate for π is $\frac{1}{n}\sum_{i=1}^{n} X_i$.

Answer 8.2: $\mathbb{E}[X_i] = \mathbb{E}[X_j]$ for all $i \neq j$.

Answer 8.3: $\mathrm{Var}[X_i] > 1$.

Answer 8.4: If $D = \{0, 1, 1, 0, 1\}$, then the likelihood of π is $\pi^3(1 - \pi)^2$.

Answer 8.5: $P(X_i = 1 \cup X_j = 0) > P(X_i = 1) + P(X_j = 0)$.

Question 9

A train travels regularly from Paris to Berlin. The train takes the northern route passing through Brussels in 60 per cent of the cases and it takes the southern route passing through Strasbourg otherwise. The train arrives Berlin later than the planned schedule in 30 per cent of the cases when it takes the northern route and in 10 per cent of the cases when it takes the southern route. Evaluate the following statements about this train.

Answer 9.1: Observing that the train is late increases the probability that it took the northern route.

Answer 9.2: Observing that the train is late increases the probability that it took the southern route.

Answer 9.3: If the train chose the northern and southern route with equal percentages, then the probability that the train took the northern route given that it arrived Berlin late would be greater than the probability that the train took the southern route given that it arrived Berlin late.

Answer 9.4: If the train is observed to arrive Berlin late, then the probability that it took the northern route is smaller than the probability that it took the southern route.

Answer 9.5: Probability that the train will arrive Berlin late is greater than the probability that it will arrive Berlin late given that it took the northern route.

Question 10

Evaluate the following statements about the $k-$nearest neighbor classifier. Assume that all the conditions other than the ones specified in the statements are equal. k always denotes only the number of nearest neighbors used for classification.

Answer 10.1: Model bias increases if the training set size grows.

Answer 10.2: Model variance increases if k grows.

<u>Answer 10.3:</u> The model has higher overfitting risk if training set size shrinks.

<u>Answer 10.4:</u> The underfitting risk of the model increases if k grows.

Answer 10.5: The training error is guaranteed to be zero for all $k > 0$.

Question 11

```
for epoch in range(20):
    for inputs_batch, outputs_batch in train_dl:
        predictions_batch = model(inputs_batch.view(-1, 5))
        loss = ((predictions_batch - outputs_batch) ** 2).mean()
        loss.backward()
        optimizer.step()
        optimizer.zero_grad()
```

Evaluate the following statements related to the Python source code given above which uses the PyTorch library.

Answer 11.1: The code is usable for the training stage of a regression task.

Answer 11.2: The input dimensionality is five.

Answer 11.3: The command `optimizer.step()` updates the gradients of the model parameters.

Answer 11.4: The command `loss.backward()` updates the learnable parameters of the model.

Answer 11.5: The code uses each data point within the set `train_dl` exactly 20 times for training

Question 12

```
model = nn.Sequential(
  nn.Linear(500, 100),
  nn.ReLU(),
  nn.Linear(500, 5),
  nn.Softmax()
)
```

Evaluate the following statements about the PyTorch source code given above. Correction: Interpret the first line as `nn.Linear(100, 500)`. Fact will be taken into account during grading.

Answer 12.1: The code builds a neural network with a single hidden layer containing 100 neurons.

Answer 12.2: The model assumes that the input data points have 100 features.

Answer 12.3: The model is designed to solve a classification task with 5 classes.

Answer 12.4: The model has more than 100000 parameters.

Answer 12.5: The model uses the sigmoid activation function between the hidden layer and the output layer.